
SGIQ: Sparse-Gradient-Informed Gradient Estimation for Variational Quantum Circuits

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Abstract

Gradient estimation is a major measurement bottleneck in training complex variational quantum circuits (VQCs) because backpropagation scaling is unattainable for general cases with unknown quantum inputs and no quantum memory. Existing gradient estimators therefore face a cost-accuracy trade-off: parameter-shift rule (PSR) incurs a cost linear in the parameter dimension, while simultaneous perturbation stochastic approximation (SPSA) uses few evaluations but can suffer from dimension-dependent variance. To improve this trade-off, we exploit settings in which loss-gradient energy concentrates on a small subset of parameters. We formalize this property through gradient-energy sparsity and derive sufficient conditions from causal-cone, commutation, and response-decay mechanisms. Building on this structure, we propose SGIQ, a Sparse-Gradient-informed Quantum gradient estimation algorithm that uses averaged SPSA estimates to rank all circuit parameters, replaces the selected coordinate estimates with PSR gradients, and retains the SPSA estimates elsewhere. Its cost is determined by the circuit’s gradient structure, rather than the parameter dimension. In ideal and finite-shot regression experiments, SGIQ uses fewer gradient circuits than PSR and attains lower normalized test mean absolute error (MAE) than SPSA baselines.

1 Introduction

Gradient estimation is a major measurement bottleneck in training variational quantum circuits (VQCs), whose measured outputs define trainable models for quantum machine learning and variational optimization [1, 2]. Unlike classical backpropagation, derivatives through a measured quantum circuit generally require additional quantum evaluations, and backpropagation scaling is unattainable for general quantum models with unknown quantum inputs and no quantum memory [3]. The parameter-shift rule (PSR) provides a standard approach to computing VQC gradients [4, 5], but its circuit cost scales linearly with the number of trainable parameters. For d trainable parameters, a mini-batch of B samples, and Q separately evaluated measurement groups per sample, full PSR requires $2BQd$ gradient-circuit evaluations per optimizer step. In contrast, simultaneous perturbation stochastic approximation (SPSA) [6] estimates all gradient coordinates from m paired perturbations using only $2BQm$ evaluations, independent of d when m is fixed. This reduction in cost, however, comes at the expense of estimation accuracy: each SPSA coordinate estimate mixes contributions from other coordinates, leading to dimension-dependent variance, while finite-shot measurements introduce additional noise. More generally, reducing the per-step directional-gradient cost does not necessarily yield a proportional reduction in total training cost, since parameter dependence can reappear through estimator variance or convergence [3, 7]. Thus, PSR and SPSA expose a fundamental cost-accuracy trade-off in quantum gradient estimation.

This trade-off raises a natural question: *must all gradient coordinates be estimated with equal accuracy?* For structured VQCs, the answer can be no. Known causal-cone, commutation, and

response-locality arguments show that some parameters can have zero or small derivatives [8–11]. These structures can cause the loss-gradient energy to concentrate on a small effective coordinate set of size $k \ll d$. In this regime, estimating every coordinate with equal precision is unnecessary, suggesting that gradient-evaluation resources should instead be allocated according to coordinate importance. We formalize this structure through *gradient-energy sparsity* and derive sufficient conditions under which exact or approximate gradient-energy concentration arises.

Building on this structure, we propose SGiQ, a Sparse-Gradient-Informed Quantum gradient estimation algorithm that combines the global coverage of SPSA with the coordinate-wise accuracy of PSR. SGiQ uses averaged SPSA estimates as a low-cost probe over all d circuit parameters and ranks the coordinates by their estimated gradient magnitudes. It then replaces the SPSA estimates on the selected top- k coordinates with PSR gradients, while retaining the SPSA estimates elsewhere. This allocation requires $2BQ(m + k)$ gradient-circuit evaluations per step, compared with $2BQd$ for full PSR, so its cost is controlled by the probing and refinement budgets $m + k$ rather than requiring PSR evaluations for all d parameters.

Existing measurement-efficient approaches explore different ways of reducing or allocating quantum gradient evaluations. Random coordinate descent evaluates randomly selected gradient coordinates [12], while Guided-SPSA combines SPSA and PSR across training samples [13]. Coyle et al. [7] develop a forward-gradient framework with adaptive measurement allocation. In contrast, SGiQ allocates accurate gradient evaluations across parameter coordinates according to their estimated importance while retaining an SPSA estimate over the full parameter space.

We summarize our contributions as follows. First, we formalize gradient-energy sparsity and derive sufficient conditions from known circuit mechanisms. Second, we introduce SGiQ, which selectively allocates accurate PSR evaluations to important gradient coordinates identified by SPSA. Third, we demonstrate gradient-energy concentration in structured VQC regression models and show, under both ideal and finite-shot simulation, that SGiQ reduces gradient-circuit cost relative to full PSR while outperforming SPSA.

2 Preliminaries

For circuit parameters $\theta = (\theta_1, \dots, \theta_d)$, define $\rho(\theta) = U(\theta)\rho_0 U^\dagger(\theta)$, where ρ_0 is the input density operator, $U(\theta)$ is the parameterized unitary, and $\rho(\theta)$ is the output density operator. For a Hermitian observable O , define the measured expectation value $E_O(\theta) = \text{Tr}[O\rho(\theta)]$. In supervised learning, measured outputs are combined through a classical readout and loss function to form the scalar mini-batch loss $L(\theta)$. The mini-batch and classical readout parameters are held fixed when estimating $\mathbf{g} = \nabla_\theta L(\theta)$.

Assume that each parameter appears in a single Pauli-generated rotation $U_j(\theta_j) = e^{-i\theta_j P_j/2}$, with $P_j^2 = I$. For the j th standard basis vector $\mathbf{e}_j \in \mathbb{R}^d$, PSR gives

$$\partial_j E_O(\theta) = \frac{1}{2} \left[E_O\left(\theta + \frac{\pi}{2}\mathbf{e}_j\right) - E_O\left(\theta - \frac{\pi}{2}\mathbf{e}_j\right) \right].$$

This identity is exact for exact expectation values [4]; finite-shot evaluations introduce sampling error. For supervised learning, PSR differentiates the measured outputs, and the chain rule yields the loss gradient. Extensions to broader generators are available [5]. For a mini-batch of B samples and Q separately evaluated measurement groups per sample, full PSR requires $2BQd$ gradient-circuit evaluations.

SPSA perturbs all parameters simultaneously [6]. For m independent Rademacher directions $\Delta^{(r)} \in \{-1, +1\}^d$ and perturbation radius $c > 0$, define

$$h_j^{(r)} = \frac{L(\theta + c\Delta^{(r)}) - L(\theta - c\Delta^{(r)})}{2c\Delta_j^{(r)}}, \quad \bar{g}_j = \frac{1}{m} \sum_{r=1}^m h_j^{(r)}. \quad (1)$$

Since $(\Delta_j^{(r)})^{-1} = \Delta_j^{(r)}$, division by $\Delta_j^{(r)}$ is equivalent to multiplication. The m paired perturbations require $2BQm$ gradient-circuit evaluations. Each shared directional difference mixes contributions from multiple gradient coordinates, while finite-shot evaluations add measurement noise.

83 3 Gradient-energy sparsity in VQCs

84 This section formalizes the gradient-energy sparsity exploited by SG1Q and derives sufficient condi-
85 tions from known causal-cone, commutation, and response-decay mechanisms [4, 5, 8–11].

86 3.1 Gradient-energy sparsity

87 Let $L : \mathbb{R}^d \rightarrow \mathbb{R}$ be a differentiable scalar loss and $g(\boldsymbol{\theta}) = \nabla L(\boldsymbol{\theta})$. For $S \subseteq [d] = \{1, \dots, d\}$, let g_S
88 retain coordinates in S and set the others to zero.

89 **Definition 1** (Gradient-energy sparsity). *For $k \in [d]$ and $\rho \in (0, 1]$, $g(\boldsymbol{\theta})$ is (k, ρ) -sparse if a set*
90 *$S \subseteq [d]$ with $|S| = k$ satisfies*

$$\|g_S(\boldsymbol{\theta})\|_2^2 \geq \rho \|g(\boldsymbol{\theta})\|_2^2. \quad (2)$$

91 When $\rho = 1$, the gradient is supported on at most k coordinates. When ρ is close to one and $k \ll d$,
92 the gradient has a small nonzero tail and is therefore compressible. This energy-based definition is
93 more appropriate than exact support sparsity for finite-shot and stochastic optimization.

94 For a nonzero gradient, let $|g|_{(1)} \geq \dots \geq |g|_{(d)}$ denote its ordered coordinate magnitudes and define

$$k_\rho(\boldsymbol{\theta}) = \min \left\{ k : \sum_{j=1}^k |g|_{(j)}^2 \geq \rho \|g(\boldsymbol{\theta})\|_2^2 \right\}. \quad (3)$$

95 This is the diagnostic quantity reported in Section 5.

96 3.2 Circuit conditions for gradient-energy sparsity

97 We next identify circuit conditions under which observable derivatives satisfy Definition 1. For an
98 observable O , define

$$E_O(\boldsymbol{\theta}) := \langle O \rangle_{\boldsymbol{\theta}} = \text{Tr}[O U(\boldsymbol{\theta}) \rho_0 U^\dagger(\boldsymbol{\theta})]. \quad (4)$$

99 Decompose the circuit around the parameterized gate $U_j(\theta_j) = e^{-i\theta_j G_j/2}$ as $U(\boldsymbol{\theta}) =$
100 $U_{>j} U_j(\theta_j) U_{<j}$, where $U_{<j}$ contains the gates before U_j , $U_{>j}$ contains those after it, and G_j is
101 the Hermitian generator. Define $O_{>j} = U_{>j}^\dagger O U_{>j}$, $\rho_j = U_j U_{<j} \rho_0 U_{<j}^\dagger U_j^\dagger$. Direct differentiation
102 gives

$$\partial_j E_O(\boldsymbol{\theta}) = \frac{i}{2} \text{Tr}(\rho_j [G_j, O_{>j}]), \quad [A, B] := AB - BA. \quad (5)$$

103 This identity links the circuit structure to individual parameter derivatives.

104 **Backward Causal Cones.** Let

$$O = \sum_{\alpha} a_{\alpha} O_{\alpha},$$

105 where each O_{α} is a local observable term. Let $\mathcal{C}_O$ contain the parameter indices whose gates lie in
106 the backward causal cone of at least one O_{α} .

107 **Proposition 1** (Causal-Cone Sparsity). *If $j \notin \mathcal{C}_O$, then*

$$\partial_j E_O(\boldsymbol{\theta}) = 0.$$

108 *Consequently, if $|\mathcal{C}_O| \leq k$, the observable-gradient vector $\nabla_{\boldsymbol{\theta}} E_O$ is $(k, 1)$ -sparse.*

109 For a shallow geometrically local circuit and a local readout, the backward cone can contain sub-
110 stantially fewer parameterized gates than the complete ansatz. Causal-cone width is also central to
111 existing analyses of gradient variance and local VQA cost functions [8, 9]. Proposition 1 uses the
112 same circuit-locality structure but only claims the exact exclusion of parameters outside the cone. Its
113 proof and a five-qubit example are given in Appendix A.1.

114 **Commutation.** Parameters outside the backward causal cone have zero derivatives, whereas param-
115 eters inside the cone are only potentially active. By Eq. (5), an inside-cone parameter still has zero
116 derivative whenever its generator commutes with the effective observable.

117 **Proposition 2** (Commutation-Induced Sparsity). *Define $\mathcal{A}_O = \{j : [G_j, O_{>j}] \neq 0\}$. For every*
118 *$j \notin \mathcal{A}_O$, $\partial_j E_O(\boldsymbol{\theta}) = 0$. Consequently, if $|\mathcal{A}_O| \leq k$, then $\nabla_{\boldsymbol{\theta}} E_O$ is $(k, 1)$ -sparse.*

This condition follows directly from the commutator representation in Eq. (5). Related commutator and Lie-algebraic representations appear in analytic quantum-gradient methods [4, 5, 10]. The proof and a diagonal Ising example are provided in Appendix A.2.

Decaying Response Tails. Exact zeros are not necessary for gradient-energy concentration. Consider an observable decomposition

$$O = \sum_X J_X h_X,$$

where X is a spatial region, h_X is supported on X , and J_X is its coefficient. Let S_j denote the qubits on which the gate parameterized by θ_j acts, and let $\text{dist}(X, S_j)$ denote the interaction-graph distance between the two supports.

Assumption 1 (Decaying Response Envelope). *There exist a constant $L_h > 0$ and a nonincreasing function $\chi : [0, \infty) \rightarrow [0, 1]$, with $\lim_{r \rightarrow \infty} \chi(r) = 0$, such that*

$$|\partial_j E_{h_X}(\boldsymbol{\theta})| \leq L_h \chi(\text{dist}(X, S_j))$$

for every parameter j and every observable term h_X .

Power-law Lieb–Robinson bounds motivate such response envelopes for Hamiltonian evolutions satisfying suitable interaction-decay conditions [11]. We use Assumption 1 only as a sufficient condition; we do not claim that it holds for every variational circuit.

Define the weighted response score

$$A_j = \sum_X |J_X| \chi(\text{dist}(X, S_j))$$

and, for a threshold $\lambda > 0$, define

$$\mathcal{S}_\lambda = \{j : A_j > \lambda\}, \quad k = |\mathcal{S}_\lambda|.$$

Proposition 3 (Compressibility from Decaying Response). *For every $j \notin \mathcal{S}_\lambda$,*

$$|\partial_j E_O(\boldsymbol{\theta})| \leq L_h \lambda.$$

Consequently,

$$\left\| (\nabla E_O)_{\mathcal{S}_\lambda^c} \right\|_2^2 \leq (d - k) L_h^2 \lambda^2.$$

If $\|(\nabla E_O)_{\mathcal{S}_\lambda}\|_2 > 0$, then

$$\frac{\|(\nabla E_O)_{\mathcal{S}_\lambda}\|_2^2}{\|\nabla E_O\|_2^2} \geq \frac{\|(\nabla E_O)_{\mathcal{S}_\lambda}\|_2^2}{\|(\nabla E_O)_{\mathcal{S}_\lambda}\|_2^2 + (d - k) L_h^2 \lambda^2}.$$

Thus, when the bounded tail is small relative to the gradient energy inside $\mathcal{S}_\lambda$, the observable gradient is (k, ρ) -sparse for ρ close to one.

The proof is given in Appendix A.3. Unlike the first two conditions, this result produces approximate concentration rather than exact zeros.

3.3 From circuit structure to loss gradients

The structural mechanisms above act first on derivatives of the measured circuit outputs. In VQC training, these outputs are combined by a classical readout and loss function, and the resulting mini-batch loss gradient is determined by the chain rule.

For sample $\mathbf{x}_i$ and measured observable O_q , define $z_{iq}(\boldsymbol{\theta}) = E_{O_q}(\boldsymbol{\theta}; \mathbf{x}_i)$. Let $\hat{y}_i = F_\phi(z_{i1}, \dots, z_{iQ})$, where ϕ denotes the classical readout parameters. For a mini-batch $\mathcal{B}$, the supervised loss is

$$\mathcal{L}_{\mathcal{B}}(\boldsymbol{\theta}, \phi) = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \ell(\hat{y}_i, \tilde{y}_i).$$

The chain rule gives

$$\partial_j \mathcal{L}_{\mathcal{B}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \frac{\partial \ell(\hat{y}_i, \tilde{y}_i)}{\partial \hat{y}_i} \sum_{q=1}^Q \frac{\partial F_\phi}{\partial z_{iq}} \partial_j E_{O_q}(\boldsymbol{\theta}; \mathbf{x}_i). \quad (6)$$

Algorithm 1 SGIQ for one mini-batch

Require: Loss L , parameters θ , SPSA directions m , number of selected coordinates k , perturbation radius c .

- 1: For $r = 1, \dots, m$, draw $\Delta^{(r)}$ and evaluate $L(\theta \pm c\Delta^{(r)})$.
 - 2: Compute $h_j^{(r)}$ for all j using Eq. (1).
 - 3: Compute $\bar{g}_j \leftarrow m^{-1} \sum_{r=1}^m h_j^{(r)}$.
 - 4: Set $s_j \leftarrow |\bar{g}_j|$ and $\mathcal{S} \leftarrow \text{TopK}(\mathbf{s}, k)$.
 - 5: Use PSR to compute g_j^{PSR} for $j \in \mathcal{S}$.
 - 6: Set $\tilde{g}_j \leftarrow g_j^{\text{PSR}}$ for $j \in \mathcal{S}$ and $\tilde{g}_j \leftarrow \bar{g}_j$ otherwise.
 - 7: **return** $\tilde{\mathbf{g}}$ to the classical optimizer.
-

149 Thus, if a common set $\mathcal{S}$, with $|\mathcal{S}| \leq k$, contains the supports of all observable derivatives in the
150 mini-batch, then $\partial_j \mathcal{L}_B = 0$ for every $j \notin \mathcal{S}$, and the mini-batch loss gradient is $(k, 1)$ -sparse. More
151 generally, Eq. (6) shows that small observable derivatives produce small loss-gradient coordinates
152 when the loss and readout derivatives are bounded. Since Eq. (6) aggregates derivatives across
153 samples and observables, the dominant coordinate set may depend on the mini-batch and current
154 parameters. Thus, the structural conditions above explain how loss-gradient concentration can arise,
155 while Sec. 5 evaluates this concentration directly using the complete mini-batch MSE gradient.

156 These conditions do not identify the dominant coordinates at each iterate, and doing so with full PSR
157 would retain its $\mathcal{O}(d)$ cost. This motivates Section 4, where averaged SPSA ranks all coordinates and
158 PSR refines only the selected top- k coordinates.

159 4 SGIQ

160 Section 3 shows that a small parameter subset can contain most loss-gradient energy in some circuits,
161 suggesting that the gradient-estimation budget need not be uniform across coordinates. The central
162 idea of SGIQ is to use averaged SPSA as a low-cost probe over all d coordinates. The resulting
163 scores identify the top- k coordinates, which are refined using PSR, while the rest retain their SPSA
164 estimates. Repeating this at each optimizer step allows the selected coordinates to change during
165 training, with a per-step cost determined by $m + k$ rather than d .

166 4.1 Gradient estimator

167 Let $\mathcal{I} = \{1, \dots, d\}$ denote the parameter index set. At each optimizer step, SGIQ computes the
168 averaged SPSA estimate $\bar{\mathbf{g}}$ using Eq. (1) and assigns each coordinate the magnitude score $s_j = |\bar{g}_j|$.
169 For $\mathbf{s} = (s_1, \dots, s_d)$, let $\text{TopK}(\mathbf{s}, k) \subseteq \mathcal{I}$ denote the indices of the k largest scores, with ties
170 resolved by a fixed rule. The selected index set is

$$\mathcal{S} = \text{TopK}(\mathbf{s}, k), \quad |\mathcal{S}| = k. \quad (7)$$

171 For each $j \in \mathcal{S}$, PSR computes a loss-gradient estimate g_j^{PSR} . As described in Section 3.3, PSR
172 first differentiates the measured circuit outputs, and the chain rule then produces the corresponding
173 mini-batch loss-gradient coordinate. The final estimator is

$$\tilde{g}_j = \begin{cases} g_j^{\text{PSR}}, & j \in \mathcal{S}, \\ \bar{g}_j, & j \notin \mathcal{S}. \end{cases} \quad (8)$$

174 Thus, PSR replaces the SPSA estimates on the selected coordinates, while the remaining coordinates
175 retain their averaged SPSA estimates.

176 4.2 Gradient-circuit cost

177 Let B denote the mini-batch size and let Q denote the number of separately evaluated observable-
178 measurement groups per sample. The quantities m and k are hyperparameters: m controls the number
179 of SPSA directions, whereas k controls the number of coordinates using PSR evaluations. We count
180 one circuit evaluation per input sample, parameter setting, and measurement group, so the per-step
181 costs are

$$N_{\text{PSR}} = 2BQd, \quad N_{\text{SPSA}} = 2BQm, \quad N_{\text{SGIQ}} = 2BQ(m + k). \quad (9)$$

PSR evaluates two shifted circuits for each of the d parameters. SPSA evaluates two circuits for each of the m perturbation directions, while SG1Q additionally evaluates two parameter-shift circuits for each of the k selected coordinates.

We consider budgets satisfying $m + k \leq d$, so that the proposed estimator uses no more gradient-circuit evaluations than full PSR, and it is strictly cheaper when $m + k < d$. Over T optimizer steps, the corresponding total costs are obtained by multiplying each expression in Eq. (9) by T .

4.3 Gradient-estimation accuracy

The average of m independent SPSA directions reduces the coordinate-mixing variance and stabilizes the magnitude scores used to identify large-gradient coordinates. Replacing the SPSA estimates on the selected set $\mathcal{S}$ with PSR gradients further improves the selected-coordinate accuracy and, under exact expectation-value evaluation, reduces the total squared gradient-estimation error relative to SPSA using the same m directions.

Assume that L has bounded third derivatives in a neighborhood of θ . A centered Taylor expansion of Eq. (1) gives

$$h_j^{(r)} = g_j + \sum_{\ell \neq j} g_\ell \Delta_\ell^{(r)} \Delta_j^{(r)} + O(c^2). \quad (10)$$

The summation is the SPSA coordinate-mixing term: the estimate of coordinate j contains signed contributions from the remaining gradient coordinates. For independent Rademacher directions, these contributions have zero mean, and

$$\mathbb{E}_\Delta[\bar{g}_j] = g_j + O(c^2), \quad \text{Var}_\Delta(\bar{g}_j) = \frac{1}{m} \sum_{\ell \neq j} g_\ell^2 + O\left(\frac{c^2}{m}\right). \quad (11)$$

Thus, averaging reduces the leading coordinate-mixing variance at rate $1/m$. While one SPSA direction gives identical score magnitudes across coordinates, multiple directions yield coordinate-dependent, increasingly stable scores. Under finite-shot evaluation, averaging also reduces measurement variation, whereas a smaller perturbation radius c increases finite-difference noise.

For any realized selected set $\mathcal{S}$, the error of Eq. (8) decomposes as

$$\|\tilde{\mathbf{g}} - \mathbf{g}\|_2^2 = \sum_{j \notin \mathcal{S}} (\bar{g}_j - g_j)^2 + \sum_{j \in \mathcal{S}} (g_j^{\text{PSR}} - g_j)^2. \quad (12)$$

Under analytical evaluation, PSR is exact on the selected coordinates, so $g_j^{\text{PSR}} = g_j$ for every $j \in \mathcal{S}$. Therefore,

$$\|\tilde{\mathbf{g}} - \mathbf{g}\|_2^2 = \|\bar{\mathbf{g}} - \mathbf{g}\|_2^2 - \sum_{j \in \mathcal{S}} (\bar{g}_j - g_j)^2 \leq \|\bar{\mathbf{g}} - \mathbf{g}\|_2^2. \quad (13)$$

Under finite-shot evaluation, the same decomposition shows that expected error decreases whenever the PSR MSE on $\mathcal{S}$ is smaller than the SPSA error it replaces. The equal-budget experiments in Sec. 5 compare this allocation against spending the same budget on extra SPSA directions.

5 Experiments

We first measure loss-gradient energy concentration in two structured VQC regression models. Then, we evaluate the cost-accuracy trade-off of SG1Q against full PSR and matched-budget SPSA under ideal and finite-shot simulation.

5.1 Experimental setup and circuits

Datasets. We use the Friedman1 and Friedman2 synthetic regression benchmarks introduced by Friedman [14], with Gaussian target noise of standard deviation 0.1. Friedman1 has five input features and Friedman2 has four. The ideal experiments use 100 training, 20 validation, and 80 test examples. The finite-shot Friedman1 experiments use 200 training, 40 validation, and 80 test examples. All estimators in each comparison use the same fixed data subsets.

Each input feature is independently min-max scaled to $[0, \pi]$, $\tilde{x}_{ij} = \pi \frac{x_{ij} - x_j^{\min}}{x_j^{\max} - x_j^{\min}}$, and the target is scaled to $[-1, 1]$, $\tilde{y}_i = \frac{2(y_i - y^{\min})}{y^{\max} - y^{\min}} - 1$. The normalization constants are computed from the full generated sample pool before the data splitting. All reported validation and test MAEs are evaluated on the normalized target scale.

Models & Training Objective. The structured Friedman2 model has eight qubits and three variational layers, giving $d = 48$ circuit parameters. As shown in Fig. 1, it measures eight local observables and uses the trainable affine readout

$$\hat{y}_i = b + \sum_{q=0}^7 w_q \langle Z_q \rangle_{\theta, \tilde{x}_i}. \quad (14)$$

The Friedman1 model uses the layered R_y - R_z and CZ-ring pattern shown in Fig. 2, related to the hardware-efficient circuits studied by Skolik et al. [15]. It has five qubits, eight layers, and $d = 80$ circuit parameters, followed by a global $Z^{\otimes 5}$ readout and a trainable scalar output factor a :

$$\hat{y}_i = a \langle Z^{\otimes 5} \rangle_{\theta, \tilde{x}_i}.$$

Both models minimize the mini-batch mean squared error

$$\mathcal{L}_B(\theta, \phi) = \frac{1}{|B|} \sum_{i \in B} (\hat{y}_i(\theta, \phi) - \tilde{y}_i)^2, \quad (15)$$

where θ denotes the circuit parameters and ϕ denotes the trainable classical parameters. No explicit regularization or weight decay is used. The quantum-gradient estimators are applied to θ , while ϕ is differentiated by automatic differentiation. PSR differentiates the measured circuit outputs and obtains the mini-batch loss gradient through the chain rule, whereas SPSA directly estimates the loss gradient by simultaneously perturbing all circuit parameters.

Both circuits instantiate the commutation mechanism described in Section 3. In the Friedman2 circuit, the final-layer parameters $\theta_{40}, \dots, \theta_{47}$ generate R_z rotations followed only by CZ gates and local Z measurements. Since CZ gates preserve the measured Z observables and the R_z generators commute with them, these eight parameters have identically zero observable derivatives and hence zero mini-batch loss gradients. The same argument applies to the five final-layer R_z parameters $\theta_{75}, \dots, \theta_{79}$ in the Friedman1 circuit with the $Z^{\otimes 5}$ readout. The measurements below quantify the broader gradient-energy concentration observed during training.

Optimization & Model Selection. All Friedman experiments use Adam with $(\beta_1, \beta_2) = (0.5, 0.9)$. Circuit parameters use learning rate 0.03, while output-scaling or linear-readout parameters use 0.1. SPSA and SG1Q use a fixed perturbation radius $c = 0.1$. The numbers of SPSA directions m and selected coordinates k are manually specified budget hyperparameters. After each epoch, we evaluate validation MAE and retain the checkpoint with the lowest value. Test MAE is reported after reloading this checkpoint and is not used for model selection.

Simulation. The ideal experiments use statevector expectation values and therefore contain no finite-shot or hardware noise. The finite-shot experiment uses a local Aer simulator with 1024 shots per circuit evaluation and no gate, readout, depolarizing, or device-calibrated noise model.

Let B be the mini-batch size, Q the number of separately evaluated circuit outputs per sample, and T the number of optimizer steps, Excluding classical computations, the total gradient circuits are

$$C_{\text{PSR}} = 2BQdT, \quad C_{\text{SPSA}} = 2BQmT, \quad C_{\text{SG1Q}} = 2BQ(m+k)T, \quad (16)$$

For Friedman1, $Q = 1$.

5.2 Gradient concentration in structured VQCs

We measure the gradient of the complete mini-batch MSE with respect to the quantum parameters. At the beginning of each run, one training mini-batch of size 20 is fixed as a probe batch. At epochs 5, 10, 15, and 20, full PSR computes

$$g = \nabla_{\theta} \mathcal{L}_{B_{\text{probe}}}(\theta, \phi).$$

These gradients are used only for diagnosis and do not affect the training updates.

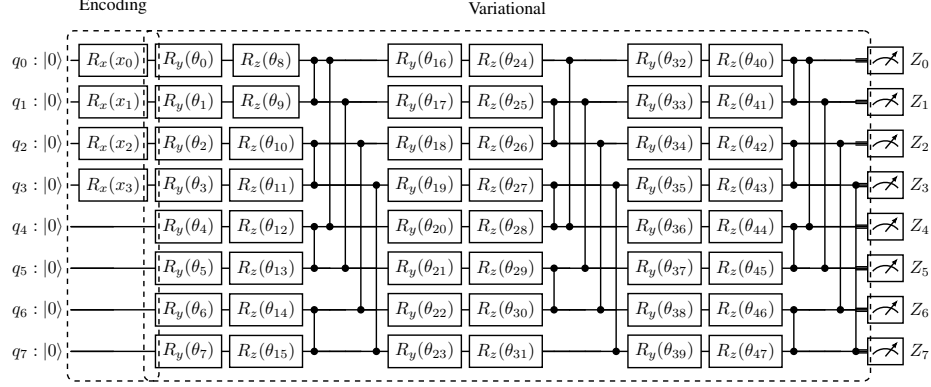


Figure 1: Sparse-local Friedman2 circuit. q_0 – q_3 encode the four features once with $R_x(x_j)$; q_4 – q_7 are auxiliary. Each of three variational layers applies trainable R_y and R_z gates to all eight qubits, followed by alternating nearest-neighbor CZ gates and feature-pair CZ gates $(0, 4)$, $(1, 5)$, $(2, 6)$, $(3, 7)$. The decoder measures the eight local observables $Z_0, \dots, Z_7$. There are $2 \times 8 \times 3 = 48$ trainable circuit angles.

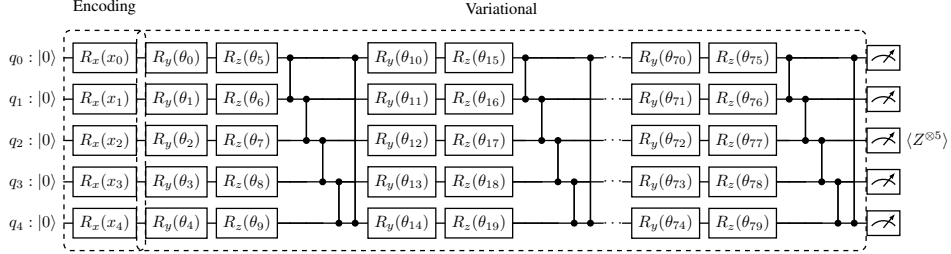


Figure 2: Friedman1 circuit. Five features are encoded once with $R_x(x_j)$. Each of eight variational layers applies trainable R_y and R_z gates to all five qubits and then the CZ ring $(0, 1)$, $(1, 2)$, $(2, 3)$, $(3, 4)$, $(4, 0)$. Layers 1, 2, and 8 are drawn; the ellipsis denotes layers 3–7. The decoder evaluates $Z_0 Z_1 Z_2 Z_3 Z_4$, and the circuit has $2 \times 5 \times 8 = 80$ trainable angles.

259 For $r \in \{1, \dots, d\}$, let $\mathcal{S}_r$ denote the index set of the r largest squared gradient coordinates. Define

$$\text{coverage}(r) = \sum_{j \in \mathcal{S}_r} g_j^2 / \sum_{j=1}^d g_j^2, \quad k_\rho = \min \{r : \text{coverage}(r) \geq \rho\}.$$

260 Thus, k_ρ is the smallest number of coordinates capturing at least a fraction ρ of the full-PSR gradient
261 energy. It is a post-hoc diagnostic and is distinct from the manually chosen algorithmic budget k .

Table 1: PSR loss-gradient energy concentration on a fixed training probe batch. k_ρ is the minimum number of coordinates capturing ρ of the total squared-gradient magnitude.

Task / Circuit	Epoch	d	k_ρ	k_ρ/d	Energy Coverage
Friedman2 / sparse-local ($\rho = 0.90$)	5	48	9	0.1875	0.9219
	10	48	12	0.2500	0.9139
	15	48	5	0.1042	0.9067
	20	48	6	0.1250	0.9133
Friedman1 / CZ ring ($\rho = 0.80$)	5	80	25	0.3125	0.8078
	10	80	29	0.3625	0.8069
	15	80	22	0.2750	0.8030
	20	80	26	0.3250	0.8082

262 Table 1 shows strong concentration in the structured Friedman2 circuit: only 5–12 of 48 parameters
263 capture at least 90% of the loss-gradient energy. Friedman1 is less concentrated but still nonuniform,

with 22–29 of 80 parameters capturing at least 80%. Since the two blocks use different target ratios, they represent two operating regimes rather than directly ranking the circuit architectures.

On structured Friedman2, full PSR obtains a test MAE of 0.3684, while SGiQ with $(m, k, c) = (10, 7, 0.1)$ obtains 0.3686. At each step, SGiQ uses $\frac{m+k}{d} = \frac{17}{48} \approx 35.4\%$ of the full-PSR gradient-circuit budget. Thus, nearly identical test MAE is achieved while evaluating substantially fewer gradient circuits.

5.3 Ideal Friedman1 comparison

Table 2: Ideal Friedman1 results with 80-parameter circuit. The entries use the same data protocol; lower normalized-scale test MAE is better. Circuit counts exclude forward and evaluation circuits.

Method	Test MAE	Cost
Full PSR	0.0964	320,000
SPSA-10 ($m = 10, c = 0.1$)	0.1526	40,000
SPSA-20 ($m = 20, c = 0.1$)	0.1555	80,000
SGiQ ($m = 10, k = 10, c = 0.1$)	0.1259	80,000

In this experiment, we use 100 training examples, batch size $B = 20$, and 20 epochs, giving $T = \frac{100}{20} \times 20 = 100$ optimizer steps. The circuit has $Q = 1$ measured output and $d = 80$ parameters. All methods are evaluated in the ideal setting. Table 2 shows that full PSR achieves the lowest test MAE, but at the highest gradient-circuit cost. SGiQ with $m = 10$ and $k = 10$ achieves substantially lower test MAE than SPSA. This shows that allocating part of the budget to selected-coordinate PSR is more effective than using the full budget for additional SPSA directions.

5.4 Finite-shot Friedman1 comparison

Table 3: Finite-shot Friedman1 results with 1024 shots per circuit evaluation. Normalized test MAE: mean $\pm$ sample standard deviation over three independent runs.

Method	Test MAE	Cost
Full PSR	0.0926 ± 0.0073	960,000
SPSA-20 ($m = 20, c = 0.1$)	0.1057 ± 0.0052	240,000
SPSA-28 ($m = 28, c = 0.1$)	0.1182 ± 0.0212	336,000
SGiQ ($m = 20, k = 8, c = 0.1$)	0.0996 ± 0.0026	336,000

The finite-shot Friedman1 experiment uses 200 training examples, batch size $B = 20$, and 30 epochs, giving $T = \frac{200}{20} \times 30 = 300$ optimizer steps. Table 3 shows that full PSR attains the lowest test MAE but uses the largest gradient budget. Notably, SGiQ uses only $28/80 = 35\%$ of the full-PSR gradient budget while achieving a lower mean test MAE than the SPSA baselines. And SGiQ achieves a lower mean test MAE and a smaller sample standard deviation than equal-budget SPSA (0.0996 ± 0.0026 versus 0.1182 ± 0.0212). The experiments connect the three components of the paper: circuit structure produces concentrated loss gradients, SGiQ allocates accurate gradient evaluations to selected coordinates, and the resulting estimator improves the cost-accuracy trade-off relative to PSR and SPSA.

6 Conclusion

Efficient quantum gradient estimation is essential for backpropagation in quantum neural networks. Motivated by concentrated loss-gradient energy, we introduced SGiQ to allocate measurement resources according to coordinate importance. Averaged SPSA provides a low-cost global probe, and PSR replaces selected coordinate estimates while SPSA estimates are retained elsewhere. Regression experiments show lower gradient-circuit cost than PSR and better predictive accuracy than matched-budget SPSA. These results support sparsity-informed gradient estimation as a practical approach to reducing the measurement cost of quantum backpropagation.

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A Structural proofs and examples

A.1 Causal-cone proof and five-qubit example

Proof of Proposition 1. For $j \notin \mathcal{C}_O$, backward propagation of each local term O_α through the gates following U_j produces an effective observable $O_{\alpha, > j}$ whose support does not intersect that of G_j . Therefore,

$$[G_j, O_{\alpha, > j}] = 0 \quad \text{for every } \alpha.$$

Using linearity and Eq. (5),

$$\partial_j E_O = \sum_{\alpha} a_{\alpha} \frac{i}{2} \text{Tr}(\rho_j [G_j, O_{\alpha, > j}]) = 0.$$

Thus, $\nabla_{\theta} E_O$ is supported on $\mathcal{C}_O$. If $|\mathcal{C}_O| \leq k$, the support can be extended to a set of size k , and Definition 1 holds with $\rho = 1$. $\square$

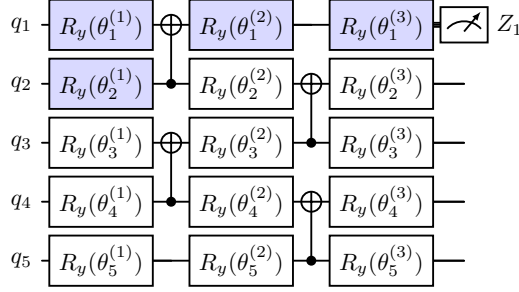


Figure 3: Five-qubit causal-cone example. The shaded rotations are the four parameters in the backward causal cone of Z_1 and are the only rotations that can affect $E_{Z_1}(\theta)$.

Example 1 (Five-qubit local readout). Consider the five-qubit circuit

$$U(\theta) = U_R^{(3)} U_E^{(2)} U_R^{(2)} U_E^{(1)} U_R^{(1)}, \quad U_R^{(\ell)} = \prod_{i=1}^5 R_y(\theta_i^{(\ell)}),$$

where

$$U_E^{(1)} = \text{CNOT}_{2 \rightarrow 1} \text{CNOT}_{4 \rightarrow 3}, \quad U_E^{(2)} = \text{CNOT}_{3 \rightarrow 2} \text{CNOT}_{5 \rightarrow 4},$$

and the measured observable is $O = Z_1$.

Propagating Z_1 backward first encounters $R_y(\theta_1^{(3)})$. Neither CNOT in $U_E^{(2)}$ expands its support, so the middle rotation layer contributes only $R_y(\theta_1^{(2)})$. Under $\text{CNOT}_{2 \rightarrow 1}$,

$$Z_1 \mapsto Z_2 Z_1,$$

and hence the first rotation layer contributes $R_y(\theta_1^{(1)})$ and $R_y(\theta_2^{(1)})$.

The backward-cone parameter set is therefore

$$\mathcal{C}_O = \{(1, 1), (1, 2), (2, 1), (3, 1)\},$$

where (ℓ, i) indexes $\theta_i^{(\ell)}$. Only four of the fifteen rotation parameters can have nonzero derivatives. Consequently,

$$\nabla_{\theta} E_{Z_1}$$

is supported on at most four coordinates and is therefore $(4, 1)$ -sparse.

A.2 Commutation proof and diagonal Ising example

Proof of Proposition 2. If $j \notin \mathcal{A}_O$, then

$$[G_j, O_{>j}] = 0.$$

Equation (5) therefore gives

$$\partial_j E_O(\theta) = \frac{i}{2} \text{Tr}(\rho_j [G_j, O_{>j}]) = 0.$$

Hence, $\nabla_{\theta} E_O$ is supported on $\mathcal{A}_O$. If $|\mathcal{A}_O| \leq k$, it satisfies Definition 1 with $\rho = 1$. $\square$

Example 2 (Diagonal Ising block). Let $\mathcal{G} = (V, E)$ be a weighted graph and define

$$O_G = \sum_{(u,v) \in E} w_{uv} Z_u Z_v, \quad U_D(\gamma) = \prod_{(u,v) \in E} e^{-i\gamma_{uv} Z_u Z_v / 2}.$$

Here, V and E are the vertex and edge sets, w_{uv} is the weight of edge (u, v) , and γ_{uv} parameterizes the corresponding diagonal gate. Consider

$$U(\beta, \gamma) = U_D(\gamma) U_M(\beta),$$

where $U_M(\beta)$ is a mixer block with r trainable parameters.

356 Every generator in U_D commutes with every term in O_C . Therefore,

$$U_D^\dagger(\gamma)O_C U_D(\gamma) = O_C$$

357 and

$$\frac{\partial E_{O_C}}{\partial \gamma_{uv}} = 0 \quad \text{for every } (u, v) \in E.$$

358 The mixer coordinates may remain active; for example,

$$[X_u, Z_u Z_v] \neq 0.$$

359 The circuit contains

$$d = r + |E|$$

360 parameters but has at most r potentially nonzero gradient coordinates. Thus,

$$\nabla E_{O_C}$$

361 is $(r, 1)$ -sparse. For a dense n -vertex graph with $|E| = \Theta(n^2)$ and $r = O(n)$, the fraction of
362 potentially active coordinates satisfies

$$\frac{r}{r + |E|} = O\left(\frac{1}{n}\right).$$

363 This example demonstrates algebraic sparsity: a parameter can have zero derivative even when its
364 gate is geometrically connected to the measured qubits.

365 A.3 Proof of the decaying-response bound

366 *Proof of Proposition 3.* By linearity of the expectation value and the triangle inequality,

$$|\partial_j E_O| = \left| \sum_X J_X \partial_j E_{h_X} \right| \leq \sum_X |J_X| |\partial_j E_{h_X}|.$$

367 Using Assumption 1,

$$|\partial_j E_O| \leq L_h \sum_X |J_X| \chi(\text{dist}(X, S_j)) = L_h A_j.$$

368 For $j \notin S_\lambda$, $A_j \leq \lambda$, and therefore

$$|\partial_j E_O| \leq L_h \lambda.$$

369 Summing this squared bound over the $d - k$ coordinates outside S_λ gives

$$\|(\nabla E_O)_{S_\lambda^c}\|_2^2 \leq (d - k) L_h^2 \lambda^2.$$

370 Finally,

$$\|\nabla E_O\|_2^2 = \|(\nabla E_O)_{S_\lambda}\|_2^2 + \|(\nabla E_O)_{S_\lambda^c}\|_2^2,$$

371 which yields

$$\frac{\|(\nabla E_O)_{S_\lambda}\|_2^2}{\|\nabla E_O\|_2^2} \geq \frac{\|(\nabla E_O)_{S_\lambda}\|_2^2}{\|(\nabla E_O)_{S_\lambda}\|_2^2 + (d - k) L_h^2 \lambda^2}.$$

372 This is the gradient-energy coverage condition in Definition 1. □

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- 704 • Depending on the country in which research is conducted, IRB approval (or equivalent)
- 705 may be required for any human subjects research. If you obtained IRB approval, you
- 706 should clearly state this in the paper.
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- 708 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
- 709 guidelines for their institution.
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- 711 applicable), such as the institution conducting the review.

712 **16. Declaration of LLM usage**

713 Question: Does the paper describe the usage of LLMs if it is an important, original, or

714 non-standard component of the core methods in this research? Note that if the LLM is used

715 only for writing, editing, or formatting purposes and does *not* impact the core methodology,

716 scientific rigor, or originality of the research, declaration is not required.

717 Answer: [N/A]

718 Justification: **[TODO]**

719 Guidelines:

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- 722 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
- 723 be described.